

ZHENGTAO GAO

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Education

Università della Svizzera italiana (USI, Lugano, Switzerland)
Master of Science in Artificial Intelligence

Sept 2024 – Present
GPA: 9.5/10

Nankai University (Tianjin, China)
Bachelor of Science (Major: Chemical Biology)

Sept 2019 – Jun 2023
GPA: 3.7/4.0 (87.98/100)

Publications

- [1] H. Jing, **Z. Gao**, S. Xu, T. Shen, Z. Peng, S. He, T. You, S. Ye, W. Lin, S. Sun. Accurate prediction of antibody function and structure using bio-inspired antibody language model. In: *Briefings in Bioinformatics* 25.4 (2024). DOI: [10.1093/bib/bbae245](https://doi.org/10.1093/bib/bbae245).

Research Experience

Cross-User Generalization and Personalization for Multimodal Smart-Glass Data

Jul 2025 – Mar 2026

↪ Research Assistant, Advisor: Dr. Martin Gjoreski

People-Centered Computing Lab, USI

- Identified inter-subject domain shift as a key challenge in multimodal smart-glass sensor data, which leads to significant performance degradation when generalizing to unseen users.
- Pretrained a ViT model using MAE on multimodal sensor data augmented with Euler angle features, and leveraged user identity as a zero-cost label to investigate its effect on cross-user generalizability.
- Fine-tuning the pretrained model with only 5% of labeled data achieves comparable or superior performance to a fully supervised baseline trained on 100% of labeled data, across two downstream tasks.

Contrastive Learning based Molecular Clustering [\[GitHub Link\]](#)

Jan 2023 – May 2023

↪ Graduation Thesis, Advisor: Prof. Wensheng Cai

Laboratory of Chemoinformatics, Nankai University

- Adapted the SwAV contrastive framework to molecular dynamics, designing a GNN to extract end-to-end representations directly from raw 3D coordinates by enforcing translational invariance via zero-centering and empirical rotational invariance through $SO(3)$ data augmentation.
- Benchmarked on alanine dipeptide trajectories, demonstrating that the learned representations effectively capture the distinct metastable states present in the classical free-energy landscape.

Bio-Inspired Antibody Language Model (BALM)

Jul 2022 – Jan 2023

↪ Research Assistant, Advisor: Prof. Siqi Sun

Institute of Intelligent Complex Systems, Fudan University

- Motivated by the limitation of general protein language models like ESM-2 in capturing antibody-specific properties due to their unique evolutionary patterns and high flexibility in antigen-binding regions.
- Designed the BALM architecture to incorporate biological inductive biases by introducing region-aware positional encodings and conservation-based masking strategies.
- Preprocessed 1B+ sequences and conducted distributed pre-training on 336M non-redundant samples, achieving SOTA performance across functional properties and structure prediction tasks.

Awards

Foundation for the Lugano Faculties Scholarship, USI Awarded to the best student in Master's Program. **2025**

Foundation for the Lugano Faculties Scholarship, USI Selected as 1 of 60 recipients via application. **2024**

Academic Excellence Scholarship, Nankai University **2019 – 2020, 2020 – 2021**

Selected Projects

Interactive Cooking Agent *Course Project, Natural Language Processing*

Autumn 2025

- Built a RAG pipeline indexing 2.2M recipes using FAISS (IVF quantization) and sentence transformers, powering semantic retrieval for a locally deployed Qwen3-8B model via Ollama.
- Designed a multi-node LangGraph agent with intent classification, context-aware memory summarization, and step-by-step interactive guidance, enabling real-time conversational cooking assistance.
- Conducted a qualitative LLM-as-judge evaluation using Gemini 3 Pro, confirming the designed agent scored 5/5 in interactive experience versus 1/5 for a naive retrieval-only baseline.

Reproduction of g-GRFs *Reproducibility Challenge, Graph Deep Learning* [\[GitHub Link\]](#)

Autumn 2025

- Reproduced all five experiments from *General Graph Random Features* (ICLR 2024), extending the original codebase which only provided a partial implementation for Experiment 1.
- Achieved $> 10\times$ speedup for Experiment 1 through Numba JIT compilation and cache optimization.

Photon Mapping Renderer (C++) *Rendering Competition, Computer Graphics* [\[GitHub Link\]](#) **Autumn 2025**

- Implemented photon mapping with KD-tree acceleration for global illumination in a C++ ray tracer.
- Invited to present at the USI Autumn Semester Project Presentations.

Monte Carlo Localization Simulator (Python) *Course Project, Robotics* [\[GitHub Link\]](#) **Spring 2025**

- Implemented a configurable Monte Carlo Localization simulator in Python, integrating differential drive kinematics, ray-casting distance sensing, and Gaussian likelihood weighting.
- Accelerated ray intersection computation by $10\times$ and reduced overall rendering time by 50% by replacing Shapely API calls with custom vectorized NumPy cross-product logic.
- Improved tracking robustness and prevented particle depletion during sharp turns by designing a hybrid Low-Variance resampling strategy paired with exponential moving average weight smoothing.

Technical Skills

Programming Languages	Python, C++, MATLAB
ML / DL	Transformers, Representation Learning, Self-supervised learning, GNN
Python Libraries	PyTorch, NumPy, pandas, Matplotlib, LangChain and LangGraph
Systems & Tools	Linux, Git, L ^A T _E X
Languages	English (fluent), Italian (basic), Chinese (native)